

SARAVANAKUMAR MURUGAN

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SUMMARY

Mechanical Engineering Graduate Student specializing in advanced thermal management, electronics cooling simulations, and data-driven hardware optimization. Proven track record conducting high-energy aerospace blast validation testing and multi-variable non-linear structural failure modeling. Proficient in high-fidelity FEA/CFD multi-physics software coupled with custom data engineering pipelines to drive hardware reliability.

EDUCATION

University of Texas at Arlington

MS in Mechanical Engineering

Expected Dec 2026

Arlington, TX

Karunya Institute of Technology and Sciences

B.Tech in Aerospace Engineering

May 2023

Coimbatore, India

EXPERIENCE

SDSC SHAR, Indian Space Research Organisation (ISRO) — Project Intern Dec 2022 – Jan 2023

- Conducted 1:12.8 scaled-down vertical hot-flow blast validation tests to evaluate the intense structural dynamic environments, thermal gradients, and acoustic profiles hitting launch pad hardware during liftoff.
- Collected, scrubbed, and filtered real-time transient data arrays using thin foil-type heat flux gauges and K-type thermocouples across top and bottom surfaces of Mobile Launch Pedestal (MLP) structures to isolate peak thermal path boundaries.
- Configured high-velocity physical tests at zero liftoff height ($L/D_e = 0$) utilizing twin Adithya solid motors to output a Mach 3.38 environment paired with a central Surya motor pushing a scaled mass flow rate of 3.715 kg/s.

SS Technologies — Engineering Intern

Nov 2021 – Jan 2022

- Operated Systems Tool Kit (STK) to architect complex orbital mission simulation models, executing high-fidelity trajectory analyses, line-of-sight (LOS) evaluations, and multi-body orbit propagation workflows.
- Applied advanced aerospace engineering principles to satellite and launch vehicle layouts, modeling orbital asset coverage passes and verifying communication linkage windows under strict mission constraints.

1Stop.ai — Engineering Intern

Sep 2021 – Nov 2021

- Completed a 3-month AI/ML program covering deep learning, CNNs, NLP, TensorFlow, and Keras; built a Celebrity Image Classifier and Fashion MNIST apparel classification model using CNNs.
- Engineered data cleaning and preprocessing pipelines using NumPy and Pandas to build model-ready inputs across multiple classification projects.

Beat The Virus Startup — Project Intern

Mar 2021 – Apr 2021

- Delivered 3 industrial AI projects under mentor evaluation: object recognition with CNN, handwritten digit classification, and a news topic classifier using NLP pipelines (tokenization, stemming, NLTK).
- Applied dense hidden layer tuning and activation function optimization to improve classification accuracy across real-world datasets.

PROJECTS & RESEARCH

270W Electronics Package Fin Heat Sink Thermal Optimization

- Modeled a comprehensive 1-D analytical thermal resistance network to determine solid-stack constraints ($R_{solid} \leq 0.104K/W$) for managing an elite 270W TDP processor load.
- Designed an optimized copper plate-fin heat sink layout in ANSYS Icepak; engineered a non-conformal structural assembly grouping the substrate, die, and heat spreader to resolve meshing priority errors and mesh regions separately.
- Analyzed temperature profiles and core velocities across 15, 30, and 40 CFM; mapped static pressure drop metrics directly across the heat sink core to hold stable die junction bounds ($77.42^{\circ}C$), hitting an optimized PUE of 1.93.

Aircraft Fleet Fuselage Lap Joint Widespread Fatigue Damage Analysis

- Conducted a forensic failure investigation on historical flight decompression ruptures (Aloha 243 & Southwest 812) to isolate non-linear failure mechanics across overlapping skin structures.
- Modeled the exact damage coupling where environmental moisture breakdown triggers crevice/pitting corrosion, creating rapid stress concentrations that manifest as Multiple-Site Fatigue Damage (MSD) around fastener paths.
- Structured a laboratory verification testing framework using tension-tension cyclic loading and an artificial marine environment block ($34.6-35PSU$, low-frequency load spectrums at $0.01-1Hz$) to predict true structural lifespans.

Non-Linear Flexible Retaining Ring FEM Structural Optimization

- Simulated non-linear multi-axial mechanical tension loading for a flexible ring clip using Abaqus CAE to isolate peak structural stress concentration zones.
- Ran a multi-stage mesh convergence validation study through 6 mesh density partition sets (scaling from 102 to 868 elements) to prove analytical grid independence and eliminate localized mesh error.
- Discovered structural yield threshold breaches on standard Carbon and Stainless Steel components, establishing Beryllium Copper as the ideal alternate alloy via normalized stress-to-yield ratio charting.

Supersonic Conceptual Interceptor Flight Sizing Synthesis

- Developed a systematic statistical script using performance metrics of high-speed aircraft to execute configuration sizing, bounding fuel, payload, and gross takeoff parameters.
- Generated complete aerodynamic performance profiles mapping zero-lift parasitic drag, supersonic wave-drag boundaries, and built a full operational $V-n$ Diagram defining extreme $+7.5G$ and $-3.75G$ limit parameters.

Laser Propulsion — Co-authored Research Publication

- Modeled thermally-limited acceleration of laser-driven light sails using Stefan-Boltzmann thermal balance equations, evaluating feasible sail area ranges across 1–2000 GW laser power inputs for c-Si, SiC, and Graphene material combinations.
- Generated Temperature-Area, Acceleration-Temperature, and Velocity-Area profiles to define mission-specific design boundaries for interplanetary laser propulsion systems. SSRN — doi:10.2139/ssrn.4410561.

TECHNICAL SKILLS

- **CAD & Solvers:** ANSYS Icepak, Abaqus CAE, SolidWorks, ANSYS (FEA/CFD), Creo, AutoCAD
- **Data & Computation:** Python (NumPy, Pandas, TensorFlow, Keras, Scikit-Learn), MATLAB, C, LaTeX
- **Core Domains:** Electronics Cooling, Simulation Engineering, Failure Reliability Engineering, Digital Image Correlation (DIC) Metrology, Geometric Dimensioning & Tolerancing (GD&T, ASME Y14.5-2018)
- **Advanced Instrumentation:** Thermomechanical Analysis (TMA), Dynamic Mechanical Analysis (DMA), Simultaneous Thermal Analysis (STA), Nanoindentation Systems